

Precept 2 Outline

ECO 202 TAs

September 17, 2026

1 Weekly check-in

1. How are things going? Questions or comments?

2 Review of practice questions

1. **(Practice 3, Question 2)** Two historical occupational groups have the following sample summaries for annual earnings, measured in thousands of dollars:

Group	Sample mean	Sample standard deviation	One worker's earnings
A	60	8	72
B	75	10	90

- (a) Calculate the sample z-score of the listed worker in each group.
 - (b) One worker earns more in raw dollars. Who has the higher standardized position within the corresponding group? Explain the difference between the two comparisons.
 - (c) In Group A, what earnings correspond to $z = -0.75$? Reverse the standardization and state the answer in thousands of dollars and in dollars.
 - (d) Show that expressing every Group A amount in dollars rather than thousands of dollars leaves the z-score unchanged.
 - (e) Does $z = 1.5$ by itself establish an exact percentile or a causal explanation for earnings? State what additional model or design information would be needed for either claim.
2. **(Practice 3, Question 4)** In a newly constructed teaching example, weekly hours for a group of full-time workers are approximated by

$$X \sim N(40, 4^2)$$

You may use the following standard Normal values:

Standard Normal quantity	Value
$\Phi(-1.5)$	0.0668
$\Phi(-1)$	0.1587
$\Phi(1)$	0.8413
$\Phi(2)$	0.9772
90th-percentile z -value	1.2816

- (a) Identify the model mean, standard deviation, and variance, with units.
- (b) Calculate and interpret $P(X < 34)$.

- (c) Calculate and interpret $P(36 < X < 44)$.
 - (d) Calculate and interpret $P(X > 48)$ using a complement.
 - (e) Find and interpret the model's 90th percentile in hours.
 - (f) Actual weekly work hours can be heaped at round numbers and may not be symmetric. Explain why the answers above are model-based approximations and name a graphical check relevant to this question.
3. **(Previous ECO 202 HW Question)** Consider the distribution of X characterized by the density curve

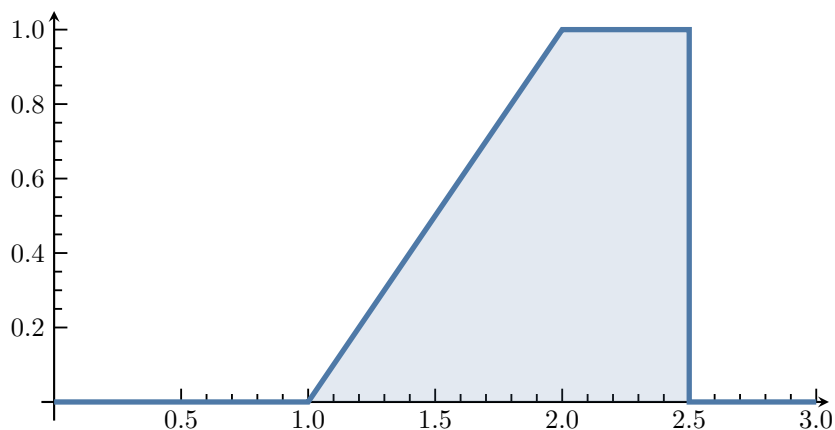
$$f(x) = \begin{cases} \frac{1}{2} & \text{for } 0 \leq x \leq 1 \\ \frac{3}{4} - \frac{1}{4}x & \text{for } 1 < x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Draw the density curve and check that the area below the curve is one.
- (b) Compute the median of this distribution.
- (c) What fraction of observations of X are larger than 2?
- (d) **(Challenge)** Provide the density curve that characterizes the distribution of $Y = X/2$.

3 Extra practice problems

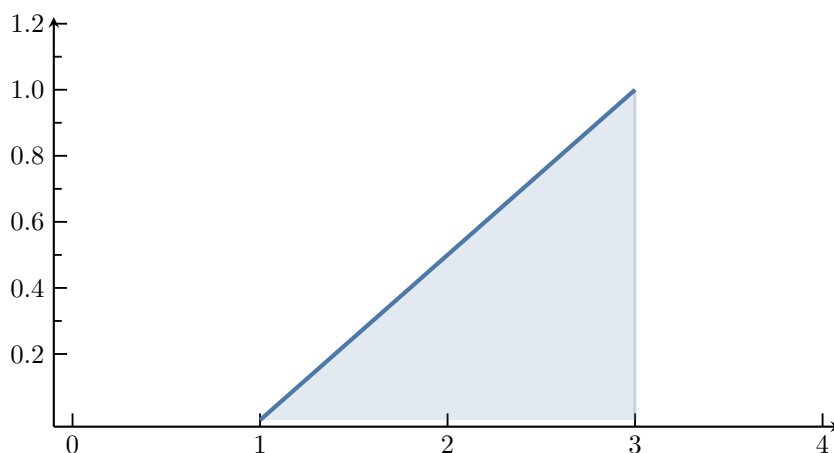
Note: Some of these are challenging!

1. Let X be the distribution described by the following density curve:



- (a) What is the median of X ?
 - (b) Is the mean smaller or larger than the median? Explain.
 - (c) What fraction of $(X + 2)/7$ is smaller than $1/2$?
2. The distribution of the SAT scores of applicants to a college is approximately distributed normal with mean 1280 and standard deviation 100.
- (a) What fraction of applicants have an SAT score between 1350-1450?
3. Suppose the weight (in carat) of a collection of 456 gemstones is approximately normal with mean 2.1 and standard deviation 0.7.
- (a) Approximately how many gemstones weigh more than 2.4 carats?

- (b) Suppose the gemstones are sorted by weight. What is the approximate weight of the 175th heaviest gemstone?
4. A quantitative variable x is observed for $n = 6$ individuals and takes on the values 2, 4, 5, 7, 8, 13.
- Sketch the boxplot of this distribution. Make sure you put numbers on the y-axis.
 - Is 13 an outlier according to the rule of thumb? Explain.
 - Find another distribution of $n = 6$ values with the same median and interquartile range as x , but with mean equal to 20.
5. The quantitative variable x is approximately normally distributed with mean 10 and standard deviation 3.
- What fraction of values of x are (approximately) larger than 12?
 - What values of x smaller than 10 are so small that they would be considered outliers according to the rule of thumb for outliers?
 - Find a normal distribution with variance smaller than 10 such that the fraction of values that are approximately larger than 12 is the same as that of x .
6. The graph below depicts the density curve of the distribution of some quantitative variable x .



- What fraction of the values of x is larger than 2?
 - Sketch, on the same graph, the density curves of x , $y = x - 2$ and $w = 2x$. Put numbers on both axes.
7. Suppose the distribution of some quantitative variable x is approximately normal with mean μ and standard deviation 4.
- Suppose $\mu = 1$. What (approximate) fraction of x is larger than 3?
 - Without assuming $\mu = 1$, consider the (approximate) fraction of x larger than 3. For what values of μ is this fraction smaller than 0.75? Explain.